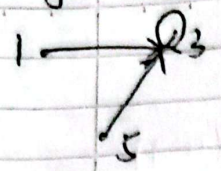
ASSIGNMENT 2.

1. i. $R = \{(1,3),(3,3),(5,3)\}$

ii. Domain $= \{1,3,5\}$

Range $= \{3\}$

iii.



iv. $(1,3) \in R$ but $(3,1) \notin R$

$(5,3) \in R$ but $(3,5) \notin R$ *

$\forall x,y \in R, (x,y) \in R \rightarrow (y,x) \notin R$

but R contains self-loop $\rightarrow (3,3)$

$\therefore$ R is not ~~assymetric~~ asymmetric.

2. $R = \{(x,y),(y,z),(x,z),(x,x),(y,y),(z,z),(y,x),(z,y),(z,x)\}$

since R is an equivalence reaction, it is reflexive, symmetric and transitive.

$\boxed{\text{REFLEXIVE}}$ $\forall x \in R, (x,x) \in R$

$\hookrightarrow (x,x),(y,y),(z,z)$.

$\boxed{\text{SYMMETRIC}}$ $\forall x,y \in R, (x,y) \in R \rightarrow (y,x) \in R$

$\hookrightarrow (x,y) \Rightarrow (y,x)$

$(y,z) \Rightarrow (z,y)$

$M_R \otimes M_R = N$

$\boxed{\text{TRANSITIVE}}$ $\forall x,y \in R, (x,y) \in R \wedge (y,z) \in R \rightarrow (x,z) \in R$

$\hookrightarrow (x,y) \in R, (y,z) \in R \rightarrow (x,z) \in R$

$(z,y) \in R, (y,x) \in R \rightarrow (z,x) \in R$

3. i. $M_R =$

	u	v	w	y
u	1	0	1	0
v	0	1	1	0
w	0	0	1	1
y	1	1	0	1

ii.

	u	v	w	y
in-degrees	2	2	3	2
out-degrees	2	2	2	3

iii. $(u,u),(v,v),(w,w),(y,y)$. It is reflexive.

$-(u,w) \in R, u \neq w, (w,u) \notin R$

$(y,v) \in R, y \neq v, (v,y) \notin R$. It is antisymmetric.

$-(u,w) \in R, (w,y) \in R \rightarrow (u,y) \notin R$. It is not transitive.

$\therefore$ since it is not transitive, R is not a partial order.

↳ wajib M_R boolean

4. If $f(x_1) = f(x_2)$

$(x_1-1)^2 = (x_2-1)^2$

$x_1 - 1 = x_2 - 1$ (It is one-to-one function)

$x_1 = x_2$

codomain $= [0,\infty] \rightarrow y \geq 0$.

$f(x) = (x-1)^2$ * general conclusion

$x = \sqrt{y} + 1$.

when $y=0, x=1. \rightarrow x \geq 1$.

(It is onto).

$\therefore$ Function f is bijection.

5. a) let $g^{-1}(y) = x$.

$y = \frac{3}{2}x - 1$.

$x = \frac{2(y+1)}{3}$.

$g^{-1}(y) = \frac{2(y+1)}{3}$.

b) $(g \circ f)(x) = gf(x)$

$= \frac{3}{2}(9x+4) - 1$.

$= \frac{27x+12}{2} - 1$.

$= \frac{27x+10}{2}$

$= \frac{1}{2}(27x+10)$.

$= \frac{27}{2}x + 5$.

5.c) $(f \circ g)(x) = f[g(x)]$

$= 9(\frac{3}{2}x - 1) + 4.$

$= \frac{27}{2}x - 5.$

d) $(f \circ g \circ g)(x) = f[g(g(x))]$

$g[g(x)] = \frac{3}{2}(\frac{3}{2}x - 1) - 1$

$= \frac{9}{4}x - \frac{3}{2} - 1$

$= \frac{9}{4}x - \frac{5}{2}$

$(f \circ g \circ g)(x) = 9(\frac{9}{4}x - \frac{5}{2}) + 4$

$= \frac{81}{4}x - \frac{45}{2} + 4$

$= \frac{81}{4}x - \frac{37}{2}$

$\cancel{= \frac{1}{4}(81x}$

$= \frac{1}{2}(\frac{81}{2}x - 37).$

6.a) $P_n = P_{n-1} + \frac{1}{4}P_{n-2}, n \geq 2$

b) $P_0 = 4, P_1 = 5$

$P_2 = P_{2-1} + \frac{1}{4}P_{2-2}$

$= P_1 + \frac{1}{4}P_0$

$= 5 + \frac{1}{4}(4)$

$= 6$

$P_3 = P_2 + \frac{1}{4}P_1$

$= 6 + \frac{1}{4}(5)$

$= 7.25$

$P_4 = P_3 + \frac{1}{4}P_2$

$= 7.25 + \frac{1}{4}(6)$

$= 8.75.$

$P_5 = P_4 + \frac{1}{4}P_3$

$= 8.75 + \frac{1}{4}(7.25)$

$= 10.5625$

$\therefore 4, 5, 6, 7.25, 8.75, 10.5625.$

7.a) input = n

output = ~~s(n)~~ s(n)

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s(n){
    If n=1
        return 2
    else
        ~~return s(n-1)~~
        return s(n-1)*s(n-1)-1
}
```

b) n=4.

s(4) | n=4
Because n≠1
return s(4-1)*s(4-1)-1 | f(4)=63 return 63

↓

s(3) | n=3
Because n≠1
return s(3-1)*s(3-1)-1 | f(3)=8 return 8

↓

s(2) | n=2
Because n≠1
return s(2-1)*s(2-1)-1 | f(2)=3 return 3

↓

s(1) | n=1
Because n=1
return 2 | f(1)=2 return 2

$\therefore f(4) = 63.$